

## QA Testing — Lessons Learned

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Version: 1.1.0 | Updated: 2026-04-30 Mac Air OBE Clean Install — March 17, 2026 Tester: Manuel Barbero | Platform: MacBook Air M4, 16GB, macOS Sequoia 15.6

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### Executive Summary

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The OBE clean install on a fresh Mac Air exposed critical gaps that would have blocked or confused any first-time user. Most issues were invisible during development because the developer machine already had all dependencies installed. The most severe issues were PDF line-wrapping breaking shell commands, Python version detection, and the start.sh demo path bug.

**Severity breakdown:** -  Showstoppers (would block install): 3 -  Significant friction (confusing, recoverable): 8 -  Polish (UX improvements): 7

**Root cause categories:** - **[ENV]** Environment assumption — dev machine had something pre-installed - **[DOC]** Documentation gap — step unclear or missing - **[BUG]** Code/config bug — actual defect in the product - **[UX]** User experience — confusing but not broken - **[PDF]** PDF rendering artifact — commands broken by line-wrapping

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### Showstopper Issues

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#### 1. Python Version Detection Failure

**What happened:** Step 7 used `python3` which resolved to the system Python 3.9 (from Xcode Command Line Tools) instead of the user-installed Python 3.14. This built a broken `.venv` that failed with:

```
TypeError: Unable to evaluate type annotation 'float | None'
```

**Root cause category:** [ENV] + [DOC] Dev machine had Python 3.13 as the default `python3`. Fresh Mac doesn't. QUICKSTART never warned that macOS system `python3` is 3.9.

**Fix applied:** Step 7 now uses auto-detect chain trying 3.14 → 3.13 → ... → 3.10.

**QUICKSTART:** Step 7 updated. **README:** Minimum Python changed to 3.10+. **TODO:** Add a preflight check script ( `scripts/check_env.sh` ) that validates Python version, PATH, and required tools before the user starts.

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## 2. PDF Line-Wrapping Breaks Shell Commands

**What happened:** The Qdrant install command in Step 5 was split across lines in the PDF. Copy-pasting wrote a broken line to `~/.zshrc`, corrupting PATH for every subsequent Terminal session. Basic commands ( `mkdir`, `ollama`, `echo` ) stopped working. **Root cause category:** [PDF] + [DOC] Multi-line commands are inherently fragile in PDF format. No warning was given.

**Fix applied:** Step 5 now uses a single `&&`-chained command that PDF cannot break. Warning added at top of QUICKSTART directing users to GitHub for copy-paste. **TODO:** All commands in QUICKSTART must be single-line or verified PDF-safe. Add automated check to CI that validates no command in QUICKSTART spans more than one line (future).

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## 3. start.sh Demo Corpus Path Bug

**What happened:** `start.sh` referenced `data/demo/demo_data` — the old path before corpus restructure. Auto-ingest failed on first run. **Root cause category:** [BUG] Corpus restructure commit updated the ingest command but missed `start.sh`.

**Fix applied:** Patched in fbfebf8, validated via `git pull` on Mac Air. **TODO:** When moving files, add a repo-wide grep for old paths as part of the commit checklist. Consider adding a test that validates `start.sh` paths exist in the repo.

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## Significant Friction Issues

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### 4. `llama3.1:8b` Not Marked as Required

**What happened:** Listed alongside optional models. Tester skipped it. Every query returned 500 error: `model 'llama3.1:8b' not found`. **Root cause category:** [DOC] **Fix applied:** Step 4 now clearly separates required vs optional. **TODO:** `start.sh` should check that `llama3.1:8b` is available before starting and print a clear error if not.

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## 5. Ollama `ollama list` Timing Issue

**What happened:** `ollama list` immediately after `brew services start` returned `could not connect to ollama server`. **Root cause category:** [DOC] Ollama needs ~3 seconds to initialize after brew starts it. **Fix applied:** Added "wait 5 seconds, then verify" to Step 3. **TODO:** `start.sh` already handles this with a wait loop — document the pattern.

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## 6. Skip Logic Not Explained

**What happened:** "Skip if already installed" confused tester — didn't know what a successful install looks like. **Root cause category:** [DOC] **Fix applied:** Every step now has explicit "if you see X — skip to next step" language. **TODO:** Consider a preflight script that checks and reports which steps can be skipped.

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## 7. # Comment Lines in Commands Cause Errors

**What happened:** Multiple `zsh: command not found: #` errors from copy-pasting blocks that included comment lines. **Root cause category:** [DOC] + [UX] Despite warnings, this kept happening because the PDF presented commands and comments as a single visual block. **Fix applied:** All `#` comments removed from all commands in QUICKSTART. **TODO:** Zero tolerance policy — enforce in doc review. Consider a linter.

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## 8. Two Pythons Coexisting Silently

**What happened:** `python3 --version` returned 3.9.6 on the same machine where `python3.14 --version` returned 3.14.0. No warning in QUICKSTART. **Root cause category:** [ENV] + [DOC] **Fix applied:** Note added explaining macOS Python version coexistence. **TODO:** preflight check script would catch this automatically.

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## 9. First Query 30-40 Second Wait With No Feedback

**What happened:** First query appeared to hang — no UI feedback during model cold load. **Root cause category:** [UX] **Fix applied:** QUICKSTART now warns: "First query may take 30-40 seconds." **TODO:** Add a loading indicator or progress message in the UI for first-query cold load. Current amber→green status pill doesn't communicate this.

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## 10. Absolute Path Required After PATH Corruption

**What happened:** After `~/.zshrc` corruption, `ollama` stopped working entirely. Required `/opt/homebrew/bin/ollama` as workaround. **Root cause category:** [PDF] (downstream of issue #2) **Fix applied:** Documented in Troubleshooting section A. **TODO:** Add `~/.zshrc` health check to preflight script.

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## 11. Port 8000 Already in Use on start.sh Re-run

**What happened:** Second `start.sh` run (after `git pull`) tried to start a second uvicorn on port 8000. Got `[Errno 48] address already in use`. Auto-ingest still completed despite the error — confusing output. **Root cause category:** [BUG] **Fix applied:** None yet — documented in troubleshooting. **TODO:** `start.sh` should check if port 8000 is occupied before starting uvicorn, and either kill the old process or skip starting a new one.

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## Polish / UX Issues

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### 12. Terminal Not Explained for Non-Technical Users

**Root cause category:** [DOC] — assumed knowledge. **Fix applied:** Step-by-step Terminal instructions, `⌘` Space shortcut explained, Dock tip added.

### 13. Progress Bar Interleaved with Qdrant Log Messages

**Root cause category:** [UX] — Qdrant and ingest run in parallel, logs mix. **TODO:** `--quiet` flag for `start.sh` (post-Beta).

### 14. Brew Version Said 4.x, Got 5.x

**Root cause category:** [DOC] — stale version number. **Fix applied:** "5.x.x or newer (current as of 2026)."

### 15. No Terminal Shortcut / Dock Tip

**Root cause category:** [DOC]. **Fix applied:** Added Dock tip and `⌘` Space instruction.

## 16. PDF vs GitHub Version Gap

**Root cause category:** [PDF]. **Fix applied:** Warning at top of QUICKSTART. **TODO:** Ship QUICKSTART.pdf with repo but watermark: "For commands, use GitHub."

## 17. HF Hub Authentication Warning

**Root cause category:** [UX] — alarming but harmless. **Fix applied:** Documented in Step 8 expected output.

## 18. UNEXPECTED CrossEncoder Key Warning

**Root cause category:** [UX] — alarming but harmless. **Fix applied:** Documented in Step 8 expected output.

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## What the Developer Machine Masked

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The following issues were invisible during development:

| Issue                                   | Why Masked                                           |
|-----------------------------------------|------------------------------------------------------|
| System Python 3.9 vs user Python 3.14   | Dev machine had 3.13 as default <code>python3</code> |
| Fresh <code>~/ .zshrc</code>            | Dev's was already correctly configured               |
| <code>llama3.1:8b</code> not downloaded | Already on dev machine                               |
| start.sh path bug                       | Demo already ingested on dev machine                 |
| First-query 40s cold load               | Model always warm on dev machine                     |
| Qdrant binary PATH issue                | Dev's <code>~/bin</code> already in PATH             |
| PDF line-wrapping                       | Dev never copy-pastes from PDF                       |

**Key lesson:** The developer environment is a terrible proxy for a fresh install. OBE on a truly clean machine should be run after every significant release.

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# Recommended Demo Questions

The validated demo benchmark set — all 14 retrieve and answer cleanly from the nine demo documents. Run `ais_bench` to execute them (source: `benchmarks/demo/demo_questions.yaml` ).

**Architecture Methodology** 1. What is QFD and how does it apply to technology architecture? 2. How do you design an IT organization around architectural principles? 3. How do architecture concepts help design organizations?

**IT Strategy & Leadership** 4. How should a CTO prioritize a three-year technology strategy? 5. What does a good technology target state look like? 6. How do you organize a large-scale IT transformation program? 7. What is the relationship between business strategy and technology strategy?

**Modern Technology** 8. What are the key considerations for cloud migration and technology modernization? 9. What are the key principles for modernizing legacy applications?

**Financial Services** 10. How do Sarbanes-Oxley and Basel II requirements shape IT governance and security architecture in financial services firms? 11. How has digitization changed financial services technology? 12. What is the role of architecture in managing technology risk in financial services?

**Agentic AI** 13. What are the key opportunities and limitations of agentic AI in financial services? 14. How should organizations embed AI into their knowledge productivity ecosystem?

# Validation Results

| Check                     | Result                   | Notes                               |
|---------------------------|--------------------------|-------------------------------------|
| Clean install from zero   | ✔ Success                | With workarounds for 3 showstoppers |
| Demo corpus auto-ingest   | ✔ After git pull fix     | fbfebf8                             |
| llama3.1:8b query         | ✔ 22s first query        | Cold load                           |
| mistral:7b query          | ✔ 18s, strong answer     | Often better than 8b                |
| Page numbers in citations | ✔ Correct                | p.20, p.16, p.5 etc.                |
| Open ↗ links              | ✔ Opens correct document | Page scroll is post-Beta            |
| git pull → instant fix    | ✔ Confirmed              | Key workflow validated              |

| Check                     | Result                                                                                      | Notes                        |
|---------------------------|---------------------------------------------------------------------------------------------|------------------------------|
| Python 3.14 compatibility |  Confirmed | Not officially tested before |

## Action Items Summary

| Item                                                         | Priority                                                                            | Owner   | Target       |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------|---------|--------------|
| preflight check script ( <code>scripts/check_env.sh</code> ) |    | Dev     | Beta         |
| start.sh: check llama3.1:8b available                        |    | Dev     | Beta         |
| start.sh: port 8000 conflict handling                        |    | Dev     | Beta         |
| start.sh: <code>--quiet</code> flag                          |    | Dev     | Post-Beta    |
| Ship QUICKSTART.pdf with GitHub watermark                    |    | Doc     | Next release |
| Repo-wide path grep on corpus moves                          |    | Process | Ongoing      |
| UI loading indicator for cold model load                     |  | Dev     | Post-Beta    |

Generated: March 17, 2026 | OBE Session — Mac Air M4 Fresh Install File:  
QA\_TESTING\_LESSONS\_LEARNED.md